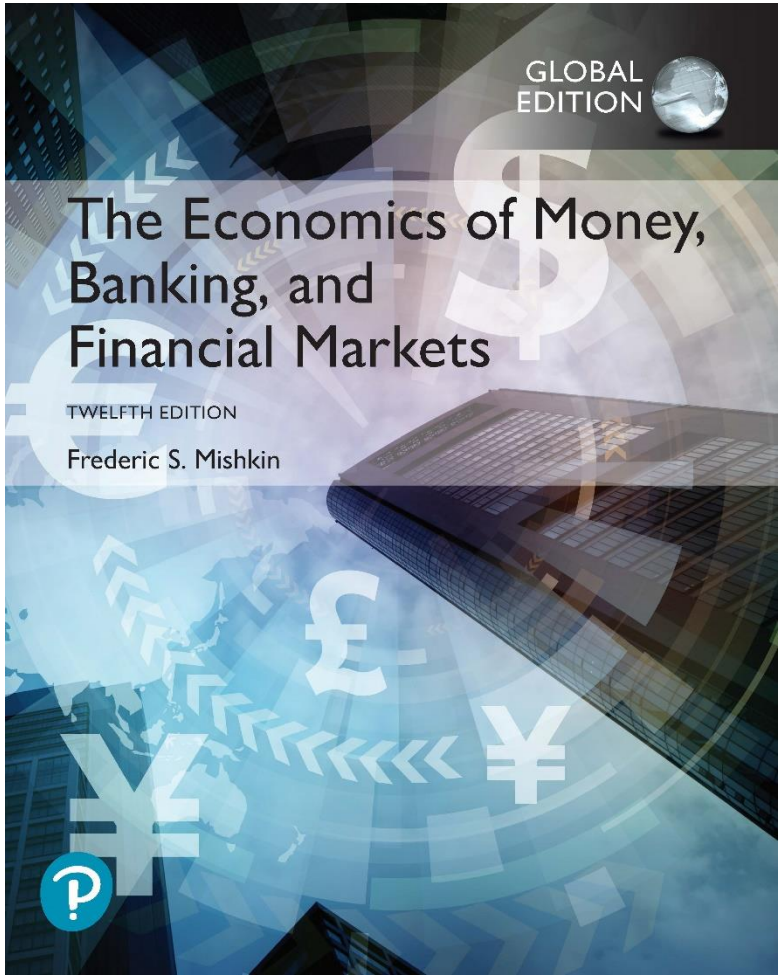
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 2

An Overview of the Financial System

Preview

- This chapter presents an overview of the study of financial markets and institutions.

Learning Objectives (1 of 2)

- Compare and contrast direct and indirect finance.
- Identify the structure and components of financial markets.
- List and describe the different types of financial market instruments.
- Recognize the international dimensions of financial markets.

Learning Objectives (2 of 2)

- Summarize the roles of transaction costs, risk sharing, and information costs as they relate to financial intermediaries.
- List and describe the different types of financial intermediaries.
- Identify the reasons for and list the types of financial market regulations.

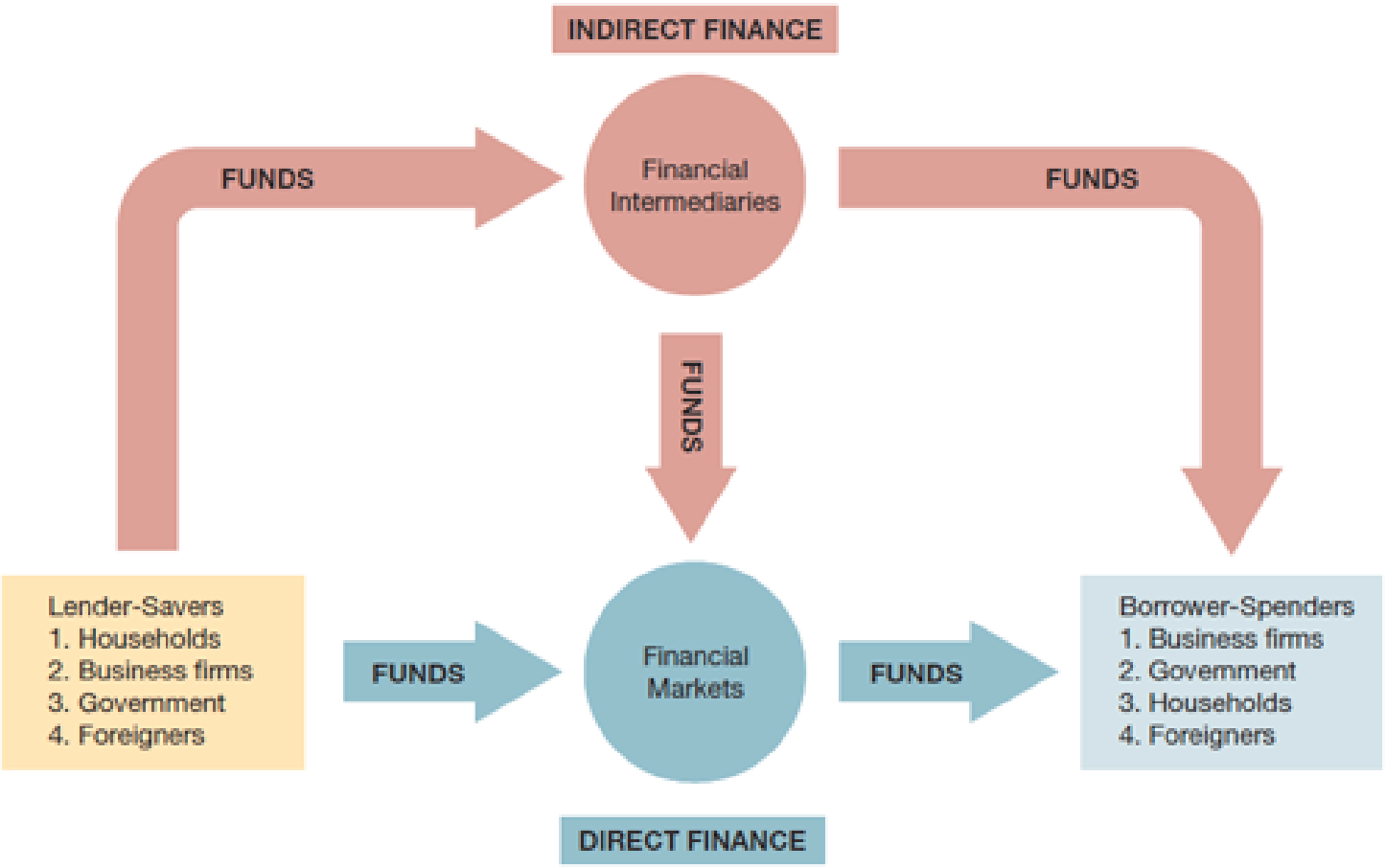
Function of Financial Markets (1 of 2)

- Performs the essential function of channeling funds from economic players that have saved surplus funds to those that have a shortage of funds
- **Direct finance**: borrowers borrow funds directly from lenders in financial markets by selling them securities

Function of Financial Markets (2 of 2)

- Promotes economic efficiency by producing an efficient allocation of **capital**, which increases production
- Directly improve the well-being of consumers by allowing them to time purchases better

Figure 1 Flows of Funds Through the Financial System



Structure of Financial Markets (1 of 2)

- Debt and Equity Markets
  - Debt instruments (**maturity**)
  - **Equities** (**dividends**)
- Primary and Secondary Markets
  - **Investment banks *underwrite*** securities in **primary markets**.
  - **Brokers**, **dealers, and investors** work in **secondary markets**.

--- interaction between primary markets and secondary markets

(price discovery, information, liquidity, corporate governance…)

Structure of Financial Markets (2 of 2)

- Exchanges and Over-the-Counter (OTC) Markets:
  - **Exchanges**: NYSE, Chicago Board of Trade
  - **OTC markets**: Foreign exchange, Federal funds
- Money and Capital Markets:
  - **Money markets** deal in short-term debt instruments
  - **Capital markets** deal in longer-term debt and equity instruments

Financial Market Instruments (1 of 2)

Table 1 Principal Money Market Instruments

	Amount ($ billions, end of year)			
Type of Instrument	**1990**	**2000**	**2010**	**2016**
U.S. Treasury bills	527	647	1,767	1,816
Negotiable bank certificates of deposit (large denominations)	547	1,053	1,923	1,727
Commercial paper	558	1,602	1,058	885
Federal funds and security repurchase agreements	372	1,197	3,598	3,778

Source: Federal Reserve Flow of Funds Accounts; http://www.federalreserve.gov

Financial Market Instruments (2 of 2)

Table 2 Principal Capital Market Instruments

	Amount ($ billions, end of year)			
Type of Instrument	**1990**	**2000**	**2010**	**2016**
Corporate stocks (market value)	3,530	17,628	23,567	38,685
Residential mortgages	2,676	5,205	10,446	10,283
Corporate bonds	1,703	4,991	10,337	12,008
U.S. government securities (marketable long-term)	2,340	3,171	7,405	12,064
U.S. government agency securities	1,446	4,345	7,598	8,531
State and local government bonds	957	1,139	2,961	3,030
Bank commercial loans	818	1,497	2,001	3,360
Consumer loans	811	1,728	2,647	3,765
Commercial and farm mortgages	838	1,276	2,450	2,850

Source: Federal Reserve Flow of Funds Accounts; http://www.federalreserve.gov

In US, GSEs such as Fannie Mae and Freddie Mac
In China, CDB bonds

Internationalization of Financial Markets

- **Foreign Bonds**: sold in a foreign country and denominated in that country's currency
- **Eurobond**: bond denominated in a currency other than that of the country in which it is sold
- **Eurocurrencies**: foreign currencies deposited in banks outside the home country
  - **Eurodollars**: U.S. dollars deposited in foreign banks outside the United States or in foreign branches of U.S. banks
- **World Stock Markets**:
  - help finance corporations in the United States and the U.S. federal government

Function of Financial Intermediaries: Indirect Finance (1 of 3)

- Lower transaction costs (time and money spent in carrying out financial transactions)
  - Economies of scale
  - Liquidity services
- Reduce the exposure of investors to risk
  - Risk sharing (asset transformation)
  - Diversification

Function of Financial Intermediaries: Indirect Finance (2 of 3)

- Deal with asymmetric information problems:
  - **Adverse Selection** (before the transaction): try to avoid selecting the risky borrower by gathering information about them
  - **Moral Hazard** (after the transaction): ensure borrower will not engage in activities that will prevent him/her to repay the loan.
    - Sign a contract with restrictive covenants.

Function of Financial Intermediaries: Indirect Finance (3 of 3)

- Conclusion:
  - Financial intermediaries allow “small” savers and borrowers to benefit from the existence of financial markets.

Types of Financial Intermediaries (1 of 5)

Table 3 Primary Assets and Liabilities of Financial Intermediaries

Type of Intermediary	Primary Liabilities (Sources of Funds)	Primary Assets (Uses of Funds)
Depository institutions (banks)		
Commercial banks	Deposits	Business and consumer loans, mortgages, U.S. government securities, and municipal bonds
Savings and loan associations	Deposits	Mortgages
Mutual savings banks	Deposits	Mortgages
Credit unions	Deposits	Consumer loans

Types of Financial Intermediaries (2 of 5)

[**Table 3** Continued]

Type of Intermediary	**Primary Liabilities (Sources of Funds)**	**Primary Assets (Uses of Funds)**
Contractual savings institutions		
Life insurance companies	Premiums from policies	Corporate bonds and mortgages
Fire and casualty insurance companies	Premiums from policies	Municipal bonds, corporate bonds and stock, and U.S. government securities
Pension funds, government retirement funds	Employer and employee contributions	Corporate bonds and stock

Types of Financial Intermediaries (3 of 5)

[**Table 3** Continued]

Type of Intermediary	Primary Liabilities (Sources of Funds)	Primary Assets (Uses of Funds)
Investment intermediaries		
Finance companies	Commercial paper, stocks, bonds	Consumer and business loans
Mutual funds	Shares	Stocks, bonds
Money market mutual funds	Shares	Money market instruments
Hedge funds	Partnership participation	Stocks, bonds, loans, foreign currencies, and many other assets

Types of Financial Intermediaries (4 of 5)

Table 4 Primary Financial Intermediaries and Value of Their Assets

	Value of Assets ($ billions, end of year)			
Type of Intermediary	**1990**	**2000**	**2010**	**2016**
Depository institutions (banks)				
Commercial banks, savings and loans, and mutual savings banks	4,744	7,687	12,821	16,834
Credit unions	217	441	876	1,238
Contractual savings institutions				
Life insurance companies	1,367	3,136	5,168	6,764
Fire and casualty insurance companies	533	866	1,361	1,908
Pension funds (private)	1,619	4,423	6,614	9,099
State and local government retirement funds	820	2,290	4,779	6,103

Types of Financial Intermediaries (5 of 5)

[**Table 4** Continued]

	Value of Assets ($ billions, end of year)			
Type of Intermediary	**1990**	**2000**	**2010**	**2016**
Investment intermediaries				
Finance companies	612	1,140	1,589	1,385
Mutual funds	608	4,435	7,873	13,616
Money market mutual funds	493	1,812	2,755	2,728

Source: Federal Reserve Flow of Funds Accounts; https://www.federalreserve.gov/releases/z1/current/data.htm, Tables L110, L114, L115, L116, L118, L120, L121, L122, L127.

Regulation of the Financial System (1 of 4)

- To increase the information available to investors:
  - Reduce adverse selection and moral hazard problems
  - Reduce insider trading (SEC)

Regulation of the Financial System (2 of 4)

- To ensure the soundness of financial intermediaries:
  - Restrictions on entry (chartering process).
  - Disclosure of information.
  - Restrictions on assets and activities (control holding of risky assets).
  - Deposit Insurance (avoid bank runs).
  - Limits on competition (mostly in the past):
    - Branching
    - Restrictions on interest rates

Regulation of the Financial System (3 of 4)

Table 5 Principal Regulatory Agencies of the U.S. Financial System

Regulatory Agency	Subject of Regulation	Nature of Regulations
Securities and Exchange Commission (SEC)	Organized exchanges and financial markets	Requires disclosure of information; restricts insider trading
Commodities Futures Trading Commission (CFTC)	Futures market exchanges	Regulates procedures for trading in futures markets
Office of the Comptroller of the Currency	Federally-chartered commercial banks and thrift institutions	Charters and examines the books of federally chartered commercial banks and thrift institutions; imposes restrictions on assets they can hold
National Credit Union Administration (NCUA)	Federally-chartered credit unions	Charters and examines the books of federally chartered credit unions and imposes restrictions on assets they can hold

Regulation of the Financial System (4 of 4)

[**Table 5** Continued]

Regulatory Agency	Subject of Regulation	Nature of Regulations
State banking and insurance commissions	State-chartered depository institutions and insurance companies	Charter and examine the books of state-chartered banks and insurance companies, impose restrictions on assets they can hold, and impose restrictions on branching
Federal Deposit Insurance Corporation (FDIC)	Commercial banks, mutual savings banks, savings and loan associations	Provides insurance of up to $250,000 for each depositor at a bank, examines the books of insured banks, and imposes restrictions on assets they can hold
Federal Reserve System	All depository institutions	Examines the books of commercial banks and systemically important financial institutions; sets reserve requirements for all banks

In China, prior to 2023, The People's Bank of China (PBC)
China Securities Regulatory Commission (CSRC）
China Banking Regulatory Commission （CBRC)
China Insurance Regulatory Commission (CIRC)
(一行三会)

In March 2023, merge into a single regulatory body